Simon Li (李冠毅)



Digital IC Design Engineer with 1+ year of hands-on experience in ASIC front-end development, including RTL design, synthesis, timing analysis, gate-level simulation (GLS), and mixed-signal integration.

Passionate about ASIC front-end design, digital IP integration, and digital interface development, with a strong interest in building high-performance and scalable digital systems.

TOEIC 890

Mobile: +886-952-765-169
E-mail: s51511830@gmail.com

Professional Experience

SiliconCore Technology Taiwan Branch

Digital IC Design Engineer **2025.10 ~ Present**

- Designed and integrated RTL modules for MicroLED display driver ICs using Verilog, contributing to ASIC front-end development including RTL implementation, synthesis, timing analysis, gate-level simulation (GLS), and tape-out support.
- Developed and executed full-chip gate-level simulation (GLS) flows using Cadence Xcelium for mixed-signal ASIC verification.
- Debugged gate-level simulation issues including X propagation, reset synchronization, timing race conditions, and netlist connectivity through waveform analysis.
- Collaborated closely with analog/schematic design, FPGA, and software teams on digital IP integration and mixed-signal hierarchy verification.
- Supported tape-out activities by resolving design issues and improving overall chip robustness through pre-silicon verification and gate-level debugging.

Technical Skills

ASIC Front-End & Digital Integration

RTL Design, Verilog, Digital IP Integration, Gate-Level Simulation (GLS), Functional Verification, Mixed-Signal Integration

Design Implementation

Logic Synthesis, Timing Analysis, Static Timing Analysis (STA), Timing Optimization, Tape-out Support

EDA Tools

Cadence Xcelium, Cadence Genus, Cadence Innovus, Synopsys Design Compiler, Synopsys IC Compiler, ModelSim, Linux

Programming

Verilog, MATLAB

Selected Projects

MicroLED Test Chip — RTL Design & Implementation

- Designed and integrated digital IP blocks, including a 256-bit multi-mode Pattern Generator and configuration register subsystem for a MicroLED test ASIC.
- Developed configuration register (CFG Register) block and control logic in Verilog for programmable chip operation.
- Integrated multiple digital IP blocks into the top-level design hierarchy and verified interface connectivity.
- Performed RTL implementation, functional verification, and logic synthesis using Cadence Genus.
- Achieved the 125 MHz timing target (WNS +3.2 ns).

MicroLED Test Chip — Full-Chip Gate-Level Verification

- Developed gate-level simulation environment for a full-custom MicroLED test chip using Cadence Xcelium.
- Diagnosed and fixed X propagation across ENGINE01 signal paths.
- Verified gate-level functionality of multiple sub-blocks.
- Built module-level testbenches and established waveform-based pass/fail verification criteria.
- Identified schematic-level design issues involving incomplete reset coverage and timing race conditions.
- Collaborated with circuit designers using annotated waveform evidence to accelerate issue resolution.
- Distinguished zero-delay GLS artifacts from functional design defects through timing analysis and hierarchical debugging.

Master's Research

Fast-Convergent Deep Reinforcement Learning Accelerator

- Developed the DRL algorithm and corresponding Verilog accelerator architecture.
- Presented at IEEE GCCE 2023, Nara, Japan.
- Completed the RTL-to-GDSII implementation flow under TSMC 90nm, including RTL verification, synthesis, APR, STA, and physical verification.
- Achieved approximately 75% energy reduction and 20% area savings compared with the baseline architecture.